

[JPJC/2025/Q3]

- 3 You are developing a locker management system for a local gym. Each locker is associated with a unique locker ID, which is an integer. These IDs are not guaranteed to be sequential or small — customers may be assigned or choose IDs such as 101, 203, 550, etc.

To efficiently store and retrieve locker access information regardless of how large or scattered the IDs are, the system uses a hash table array of fixed size **37**.

For privacy and storage efficiency, all codewords are first encrypted using a Caesar cipher (+3 shift) and then stored in the hash table.

However, since multiple locker IDs may hash to the same index, your system must use **linear probing** to resolve collisions.

Task 3.1

Write a function `encrypt(code)` to:

- encrypt the codeword `code` using a Caesar cipher with +3 shift which wraps around after 'z' (e.g., 'z' → 'c')
 - return the encrypted string.
- [5]

Task 3.2

Write a function `hashfunction(locker_id)` to:

- implement a hash function using $\text{hash_index} \leftarrow \text{locker_id} \bmod 37$
 - return the hash index.
- [1]

Task 3.3

A locker record structure is used to store a locker ID and an encrypted codeword.

Write a function `add(locker_id, hta)` to:

- get user to input a valid codeword (only lowercase alphabetic characters are allowed)
 - use your function from **Task 3.1** to encrypt the codeword
 - store the locker record (i.e.: `locker_id` and the encrypt codeword in the hash table array `hta`)
 - return the updated hash table array.
- [8]

Task 3.4

Write a procedure `search(hta)` to:

- get the user to enter his/ her locker ID and codeword
 - use your function from **Task 3.1** to encrypt the codeword
 - perform a hash table search
 - display "LOCKER NOT FOUND" if the locker ID does not exist in the hash table
 - display "INCORRECT CODE" if the locker ID is found but the encrypted codeword in the locker record is not a match
 - display "LOCKER IS OPEN" if the locker ID is found and the encrypted codeword in the locker record is a match
- [6]

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Task 3.5

Write a program to:

- initialise a hash table array with `None` as the NULL values
- add locker (locker ID 36 and user input codeword "jurongpioneer")
- add locker (locker ID 73 and user input codeword "computing")
- search locker (user input locker ID 73 and codeword "computing")
- search locker (user input locker ID 74 and codeword "computing").

Test your program and show the output.

[4]

Save your Jupyter Notebook for Task 3.